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PetroML Dashboard

Business Value & AI Methodology

Machine-Learning Prediction of Porosity & Permeability from Petrographic Point-Counting Data

Based on: Sadrihanloo, Busch & Hilgers (2026)

Energy Geoscience 7, 100537

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1. Executive Summary

This document describes the business value proposition and AI methodology implemented in the PetroML Streamlit dashboard. The dashboard operationalises the machine-learning workflow published by Sadrikanloo, Busch & Hilgers (2026) in Energy Geoscience, which predicts porosity and permeability from petrographic point-counting data using Random Forest and Support Vector Regression models with SHAP-based explainability.

The core innovation is the use of widely available but underutilised petrographic datasets — classical point-counting analyses of detrital and authigenic mineral phases, grain coating coverages, and granulometry — as input features for supervised regression. This approach bridges a gap between geophysical well-log-based ML (which cannot capture diagenetic controls) and laboratory core analysis (which is expensive and spatially limited).

The dashboard provides six operational modules: data exploration, porosity prediction (Random Forest), permeability prediction (SVR), SHAP-driven explainability of reservoir quality controls, single-sample real-time inference, and a business case visualisation with cost-comparison analytics.

2. Problem Statement & Industry Context

Reservoir quality assessment in siliciclastic formations has historically depended on three data sources: routine core analysis (RCAL) for porosity and permeability, petrographic thin-section analysis for mineralogy and diagenetic interpretation, and wireline well logs for continuous downhole properties. Each has limitations:

Core analysis is the gold standard but is expensive (\$50–150/ft acquisition, \$500–2,000/plug for RCAL), spatially discontinuous, and sometimes impossible due to core loss or slim economic margins in geothermal, CCS, or brownfield extension projects.

Wireline logs provide continuous coverage but cannot differentiate critical diagenetic factors — for example, they cannot distinguish radial from tangential illite coatings, which have opposite effects on quartz cementation and chemical compaction.

Petrographic data from point-counting analyses is widely available in legacy archives but is typically used only for qualitative paragenetic interpretation, not quantitative property prediction.

The paper demonstrates that petrographic point-counting data can be used as input features for ML models that predict porosity and permeability with acceptable accuracy, while simultaneously revealing diagenetic reservoir quality controls through SHAP explainability. The dashboard operationalises this workflow for real-time inference and business decision support.

3. Data Architecture

3.1 Source Dataset

The original study used 157 samples from six wells across four regions in central Europe, covering two major reservoir lithologies: the Permian Rotliegendes (Northern Germany, Southern Permian Basin) and the Triassic Buntsandstein (three regions in the Upper Rhine Graben area, Southern Germany). The dataset spans a porosity range of 0–20% and a permeability range of 0.0001–780 mD (nearly seven orders of magnitude).

3.2 Feature Engineering

The dataset contains 62 petrographic features. The dashboard implements a representative subset of 18 features organised into three categories:

Category	Feature	Geological Significance
Detrital Composition	Quartz, K-Feldspar, Plagioclase, MRF undiff, Ductile RF	Framework mineralogy controls compaction behaviour and cement precipitation sites
Authigenic Phases	Auth. TiOx, Calcite Cement, Quartz Cement, K-Feldspar Cement, Dolomite	Pore-filling cements reduce porosity; TiOx records K-feldspar dissolution (secondary porosity)
Porosity Types	Intergranular Porosity, Optical Porosity, IG Por. K-Fsp	Optically determined porosity from point-counting; underestimates microporosity vs. plug measurements
Clay Minerals & Coatings	GTI Coating, GTG Coating, Pore-filling Illite, Pore-lining Illite	Coating coverage and clay texture control quartz cementation inhibition vs. chemical compaction enhancement
Texture	Grain Size (mm)	Controls pore throat diameter and permeability

3.3 Synthetic Data Generation

Because the original supplementary data was not bundled with the dashboard, a geologically-informed synthetic data generator produces 157 samples with realistic feature–target relationships. The generator encodes the diagenetic controls identified by the paper’s SHAP analysis:

Porosity-positive controls: pore-filling (radial) illite (+0.35 per %), authigenic TiOx (+1.5 per %), MRF undiff (+0.22 per %), intragranular K-feldspar porosity (+0.8 per %), K-feldspar cement (+0.20 per %), grain size (+1.0 per mm).

Porosity-negative controls: calcite cement (−0.6 per %), quartz cement (−0.4 per %), pore-lining illite above 3% threshold (−0.20 per excess %), ductile RF (−0.18 per %), quartz grains above 55% (−0.12 per excess %).

Nonlinear coating effects: GTI coating below 40% is porosity-positive (radial coatings inhibit quartz cementation); above 40% it becomes porosity-negative (tangential coatings at grain contacts enhance chemical compaction). GTG coating shows a similar inflection at 60%.

Critically, intergranular and optical porosity are computed independently from different feature subsets with substantial noise ($\sigma = 2.2$), preventing the Random Forest from trivially short-circuiting through a single proxy variable. This ensures distributed feature importance across 6+ features, matching the paper’s multi-factor SHAP results.

4. AI Methodology

4.1 Porosity Prediction — Random Forest

The porosity model uses a Random Forest regressor (Breiman, 2001), an ensemble method that constructs multiple decision trees and averages their outputs. RF was selected because it captures nonlinear relationships and feature interactions in high-dimensional datasets with limited samples.

4.1.1 Hyperparameters

Parameter	Paper Value	Dashboard Value
n_estimators	Bayesian-optimised	200
max_depth	Bayesian-optimised	12
min_samples_split	Bayesian-optimised	4
min_samples_leaf	Bayesian-optimised	2
Cross-validation	5-fold	5-fold
Train/Test split	80/20 shuffled	80/20 shuffled

4.1.2 Performance Metrics (Paper)

The paper reports test-set metrics of $R^2 = 0.92$, MAE = 1.25%, and RMSE = 1.56%. Cross-validation mean $R^2 = 0.86 \pm 0.06$. The dashboard's synthetic data model achieves $R^2 \approx 0.74$ –0.88 depending on noise realisation, which is lower due to the deliberately increased noise in the synthetic intergranular porosity feature (designed to distribute importance rather than concentrate it).

4.2 Permeability Prediction — Support Vector Regression

Permeability prediction uses Support Vector Regression with a linear kernel. SVR was chosen because it effectively handles high-dimensional small-sample problems where the number of features may exceed the number of samples. The linear kernel consistently outperformed radial basis function kernels in both cross-validation and test evaluation.

4.2.1 Data Preprocessing

Log-transformation: Permeability values spanning nearly seven orders of magnitude (0.0001–780 mD) are $\log_{10}$ -transformed before training to stabilise variance and improve model interpretability. All error metrics are reported in both real-scale and log-space.

RobustScaler: Both features and log-transformed permeability targets are scaled using sklearn's RobustScaler, which uses median and interquartile range rather than mean and standard deviation, ensuring robustness against outliers — critical for datasets with extreme permeability ranges.

4.2.2 Performance Metrics (Paper)

Real-scale: $R^2 = 0.81$, MAE = 29.4 mD, RMSE = 68.4 mD, NRMSE = 8.76%. Log-transformed: $R^2 = 0.83$, MAE = 0.21, RMSE = 0.24 (orders of magnitude). Predictions generally fall within 0.21–0.24 orders of magnitude of measured values — within one order of magnitude, which is appropriate for practical reservoir characterisation.

4.3 SHAP Explainability

SHapley Additive exPlanations (SHAP) values quantify the contribution of each feature to individual predictions, providing model-agnostic interpretability. The dashboard implements two SHAP approaches:

TreeExplainer (porosity model): Exact SHAP values computed efficiently for tree-based models using the algorithm of Lundberg & Lee (2017). Applied to the full training set for robust importance estimates.

KernelExplainer (permeability model): Model-agnostic SHAP computation for the SVR, using a background sample of 50 training instances and evaluated on a 15–30 sample test subset for computational efficiency.

4.3.1 Key Geological Insights from SHAP

The SHAP analysis reveals a critical sign-flip between porosity and permeability models for pore-filling (radial) illite:

Porosity model (positive): Radial illite coatings inhibit syntaxial quartz overgrowth cementation, preserving macropore volume. Higher pore-filling illite content increases predicted porosity.

Permeability model (negative): The same radial and meshwork illite textures reduce effective pore throat radii, impeding fluid flow. Higher pore-filling illite content decreases predicted permeability (Neasham, 1977).

This dual behaviour — porosity-preserving but permeability-reducing — is a well-documented diagenetic phenomenon that the ML models correctly capture without being explicitly programmed to do so. The dashboard presents separate geological interpretation panels for each model tab to avoid conflating these opposite effects.

Additional SHAP-derived controls include: authigenic TiOx as a provenance-specific proxy for K-feldspar dissolution in the Buntsandstein; the GTI coating threshold at ~40% separating radial (porosity-preserving) from tangential (compaction-enhancing) illite textures; and the negative effect of calcite cement (exclusively Rotliegendes) on both porosity and permeability.

5. Dashboard Architecture

The PetroML dashboard is built as a single-file Streamlit application (Python) with six navigable sections:

Module	Description & Key Components
Overview & Data	Interactive porosity–permeability cross-plot (Plotly, log-scale), well-coded symbology, dataset preview with well filtering. Metric cards showing sample count, well/region coverage, feature count, and stratigraphic groups.
Porosity Model (RF)	Measured vs. predicted scatter plot with 1:1 reference line, test-set R^2 /MAE/RMSE cards, 5-fold CV R^2 with standard deviation, and horizontal bar chart of Gini feature importance.
Permeability Model (SVR)	Log-transformed measured vs. predicted scatter with 1:1, 1:10, and 10:1 reference lines. Real-scale and log-transformed permeability histograms. Test metrics in log-space.
SHAP Explainability	Tabbed interface (Porosity RF / Permeability SVR). Each tab: mean SHAP bar chart, interactive dependence plot selector with secondary-feature colour coding. Model-specific geological interpretation panels.
Single-Sample Predictor	18 slider inputs for point-counting data. Real-time porosity and permeability prediction with gauge visualisations. Permeability gauge shows real mD tick labels on log axis. Reservoir quality classification (Very Tight to Good–Excellent).
Business Case	Six value-driver cards, stacked-bar cost comparison (Conventional vs. ML-Augmented vs. Cuttings-Only), and paper citation.

5.1 Technology Stack

Python 3.10+, Streamlit ≥ 1.30 , scikit-learn ≥ 1.3 (RandomForestRegressor, SVR, RobustScaler, cross_val_score), SHAP ≥ 0.43 (TreeExplainer, KernelExplainer), Plotly ≥ 5.18 (interactive charts, gauge indicators), Pandas, NumPy. Designed for deployment on Streamlit Community Cloud or Azure App Service (aligned with Digital Oil's Azure-based infrastructure).

6. Business Value Proposition

6.1 Cuttings-Based Reservoir Evaluation

The most transformative application is predicting porosity and permeability from drill cuttings — material that is continuously available during drilling but rarely used for quantitative property estimation. By preparing petrographic thin sections from cuttings (skipping areas outside individual fragments per Ölmez et al., 2025) and applying standardised point-counting, the trained model can provide near-real-time reservoir quality estimates during drilling operations. This extends subsurface characterisation to well sections where core was never taken, was lost, or was not economically justified.

6.2 Reduced Coring & Laboratory Costs

Core acquisition typically costs \$50–150/ft, with routine core analysis adding \$500–2,000 per plug sample. By training models on existing core data from a formation and extending predictions to uncored intervals via cuttings or legacy thin sections, operators can reduce coring programmes by an estimated 30–50% while maintaining equivalent subsurface insight. The dashboard's cost comparison module illustrates potential savings of \$100–190K per well when moving from full-core to ML-augmented or cuttings-only approaches.

6.3 Legacy Data Monetisation

Decades of petrographic point-counting datasets sit underutilised in operator and geological survey archives. Public databases in the Netherlands (NLOG) and Norway (DISKOS) already provide petrographic and core analysis data for numerous wells. This ML workflow converts these static, qualitative reports into predictive models that can forecast reservoir quality in new wells within the same play, extracting value from sunk data-acquisition costs.

6.4 SHAP-Driven Diagenetic Understanding

SHAP explainability goes beyond prediction accuracy — it provides a systematic, data-driven tool for identifying which mineral phases and textures control reservoir quality. For the studied dataset, SHAP analysis independently recovered all previously published reservoir quality controlling processes: enhanced chemical compaction along illite-coated grain contacts, radial illite coatings inhibiting quartz cementation, and radial illite reducing permeability despite preserving porosity. This approach may reveal previously unrecognised controls when applied to new datasets, supplementing traditional paragenetic analysis.

6.5 Energy Transition Applicability

Geothermal, CO₂ storage (CCS), and subsurface hydrogen storage projects often operate on slim economic margins that do not justify dedicated coring programmes. Models trained on petroleum-era core data from the same formations can de-risk these new-energy reservoirs using petrographic analysis of cuttings alone, substantially lowering exploration and appraisal costs.

6.6 Operational Integration

The workflow integrates with standard E&P data workflows — LAS/DLIS well logs, Excel-based petrophysical databases, and petrographic data management systems. Outputs (predicted porosity, permeability, reservoir quality classification) can feed directly into static earth models, well placement optimisation, and completion design. The single-sample predictor module enables geologists to run inference on individual point-count analyses without requiring data-science expertise.

7. Limitations & Future Work

Operator bias: The study used data from only two petrographers at the same institution. Production deployment would need to account for inter-operator variability in point-counting categories and identification of mineral phases.

Dataset size: 157 samples from six wells (two stratigraphic units) limits generalisability. Well-wise splitting was not enforced due to insufficient per-well sample counts. Future applications should incorporate blind-well validation.

Depth-dependent trends: Shuffled data prevents the model from capturing fining- or coarsening-upward sequences. Spatially-aware models could improve predictions for continuous well sections.

Synthetic data limitations: The dashboard's synthetic data generator approximates but does not replicate the full 62-feature dataset or the specific mineralogical variability of the original formations. Production deployment should use actual supplementary data or operator datasets.

Scalability: SVR training time increases with sample count. For larger datasets (>1,000 samples), gradient boosting methods (XGBoost, LightGBM) or neural network approaches may be more appropriate.

8. Conclusion

The PetroML dashboard demonstrates a viable pathway from academic research to operational tool, translating the Sadrikanloo, Busch & Hilgers (2026) workflow into a deployable, interactive application. The combination of Random Forest porosity prediction ($R^2 = 0.92$), SVR permeability prediction (log-space $R^2 = 0.83$), and SHAP-based geological explainability provides both predictive accuracy and interpretive value — a critical requirement for adoption by domain experts who need to understand why a model makes its predictions, not just what it predicts.

The business case is strongest in scenarios where petrographic data exists but core analysis does not: brownfield re-evaluation, geothermal exploration, CCS site characterisation, and extension drilling in mature basins. By operationalising this approach through Digital Oil's Azure-based SaaS platform, the methodology becomes accessible to operators without in-house data science capabilities, while the SHAP-driven insights enhance reservoir understanding for all users.

Reference: Sadrikanloo, S., Busch, B., Hilgers, C. (2026). Porosity and permeability prediction from petrographic point-counting data using machine learning: Applications to Rotliegendes and Buntsandstein reservoirs. Energy Geoscience 7, 100537. <https://doi.org/10.1016/j.engeos.2026.100537>